

The Impact of Reward Shaping on PPO Agent Performance in a Physics-Based Environment

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Figure 1: Experimental workflow for comparing baseline and aggressive reward-function variants in the RLGym training environment.

Abstract

This paper investigates how different reward functions affect the training performance of reinforcement learning agents in Rocket League using the RLGym framework. Rather than comparing different reinforcement learning algorithms, the project compares reward design choices under a consistent PPO-based training setup.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; *Machine learning*; *Simulation environments*.

Keywords

reinforcement learning, reward functions, RLGym, Rocket League, machine learning, PPO

1 Introduction

Rocket League is a competitive video game that combines soccer-like objectives with vehicle-based movement and physics. Because the game requires positioning, timing, ball control, and decision-making, it provides an interesting environment for reinforcement learning. In this project, I train an autonomous Rocket League agent using reinforcement learning with Proximal Policy Optimization (PPO).

The project uses RLGym [2], a Python framework for creating reinforcement learning environments. RLGym can be used with RocketSim [4], a headless simulator designed to run Rocket League simulations efficiently. This allows an agent to collect many training steps without needing to render the full game.

The goal of this project is to evaluate how different reward functions affect the agent’s learning behavior. Instead of comparing several different reinforcement learning algorithms, this project keeps PPO constant and changes the reward structure between experiments. This approach allows the project to focus on how reward design influences training performance, stability, and reward-optimization patterns.

2 Background and Literature Review

Reinforcement learning is a machine learning approach in which an agent learns by interacting with an environment and receiving rewards based on its actions. Instead of learning from labeled examples, the agent improves by exploring possible actions and receiving feedback from the environment. This makes reinforcement learning useful for sequential decision-making problems such as robotics, gaming, and simulated control tasks.

Proximal Policy Optimization (PPO) was introduced by Schulman et al. as a policy gradient method designed to improve training stability while remaining simpler to implement than earlier trust-region methods [3]. PPO limits how much the policy can change during each update, which helps reduce the chance that a single update will cause the agent’s behavior to collapse. Because of this balance between performance, stability, and implementation simplicity, PPO has become a common baseline algorithm for deep reinforcement learning experiments.

Reward shaping is another important concept in reinforcement learning. Reward shaping has been studied as a way to modify the reward signal while preserving the intended learning objective [1]. In many environments, the final objective may be too sparse to guide learning effectively. For example, if an agent is only rewarded for scoring a goal, it may struggle early in training because goals are rare during random exploration. Reward shaping attempts to solve this problem by giving the agent intermediate rewards for behaviors that are expected to support the final task objective. In this project, the offensive, ball-chasing, and balanced reward functions represent different reward shaping strategies. I explore a baseline reward shape, and then a more aggressive version of each strategy.

Simulation environments are often used in reinforcement learning because they allow agents to collect large amounts of experience without interacting with the real world. Rocket League is a useful simulated environment because it combines continuous movement, physics-based control, opponent interaction, and long-term strategy. This project uses RLGym and RocketSim to train agents in a

Rocket League-like environment [2, 4]. These tools make it possible to run large numbers of training steps efficiently while testing how different reward designs affect learning behavior.

3 Methods

The three reward approaches tested in this project were an offensive reward function, a ball-chasing reward function, and a balanced reward function. Each reward function was designed to encourage a different reward-optimization objective within the Rocket League environment. The offensive reward function emphasized actions that contributed to scoring opportunities, while the ball-chasing reward function rewarded frequent movement toward and interaction with the ball. The balanced reward function combined multiple objectives in an attempt to produce a more complete style of play. After training the baseline versions of these reward functions, more aggressive variants were created by preserving the same overall reward structure but increasing the reward weights associated with the target behaviors. This design allowed the project to compare whether increasing reward intensity, without changing the underlying behavioral objective, improved learning performance or introduced additional training instability.

3.1 Offensive Reward Function

The offensive reward function was designed to encourage behaviors that are associated with attacking play and creating scoring opportunities. Rather than directly rewarding the agent for scoring a goal, this reward function emphasized proxy behaviors that were expected to support offense. These included actions such as moving toward the ball, interacting with the ball, and directing play toward the opponent's side of the field or goal area. The goal of this reward structure was to produce an agent that played more aggressively and prioritized forward pressure.

A key limitation of this reward function is that it did not include an explicit reward for scoring goals. As a result, the agent was not directly optimized for the true objective of Rocket League, which is to outscore the opponent. Instead, the model learned from intermediate signals that were assumed to be related to offensive success. This distinction is important because an agent can accumulate reward by performing offensive-looking behaviors without necessarily converting those behaviors into goals. For example, the agent may learn to chase the ball, take touches, or move the ball forward while still failing to develop effective finishing behavior.

The more aggressive offensive variant used the same general reward structure but increased the weights on the offensive behaviors. This was intended to test whether stronger reinforcement of attacking proxies would accelerate learning or produce a more offense-oriented policy. However, because goal scoring was not directly rewarded, the aggressive variant may also amplify reward-seeking behavior that appears offensive but does not always improve actual gameplay performance. Therefore, the offensive reward results should be interpreted as measuring improvement in shaped offensive behavior rather than direct scoring ability.

3.2 Ball-Chasing Reward Function

The ball-chasing reward function was designed to encourage the agent to stay active and pursue the ball as frequently as possible.

This approach emphasized movement toward the ball, close interaction with the ball, and frequent ball touches. The purpose of this reward function was to produce a highly active agent that consistently applied pressure and remained involved in the play. Compared with the offensive and balanced reward functions, the ball-chasing reward was more specialized because it focused primarily on proximity and interaction rather than broader strategic decision-making.

This reward function represents a simple but useful form of reward shaping. In Rocket League, maintaining pressure on the ball is often important because it can prevent the opponent from gaining possession and can create opportunities for offensive plays. However, ball chasing is not always strategically optimal. A player that constantly moves toward the ball may overcommit, ignore defensive positioning, or fail to rotate properly. Therefore, while the ball-chasing reward can encourage useful activity, it may also teach the agent to prioritize constant engagement over smarter long-term positioning.

A key limitation of the ball-chasing reward function is that it did not directly reward goals, saves, or match outcomes. As a result, the agent was not explicitly trained to win games, score goals, or prevent the opponent from scoring. Instead, the model was trained using proxy signals that assumed frequent ball pressure would lead to better gameplay. This means the agent could receive high reward for chasing and touching the ball even if those actions did not lead to successful offensive or defensive outcomes. In this sense, the ball-chasing reward is useful for developing basic activity and engagement, but it does not fully capture the strategic objectives of Rocket League.

The more aggressive ball-chasing variant used the same general reward structure as the baseline ball-chasing reward function but increased the reward weights associated with the target behaviors. This was intended to test whether stronger reinforcement of ball pursuit and interaction would improve learning speed and produce a more active policy. The ball-chasing reward function therefore served as a focused test of whether heavily rewarding simple, frequent interactions with the ball could accelerate learning, while also highlighting the risk that agents may learn activity without strategy.

3.3 Balanced Reward Function

The balanced reward function was designed to encourage a more complete style of gameplay by combining multiple reward signals into a single training objective. Instead of focusing only on attacking behavior or only on chasing the ball, the balanced reward attempted to reward several behaviors that are useful in Rocket League. These included maintaining interaction with the ball, positioning in ways that supported offensive pressure, and encouraging movement patterns that were expected to produce more general gameplay competence. The purpose of this reward function was to create an agent that was less specialized than the offensive or ball-chasing agents.

Unlike a highly targeted reward function, the balanced approach attempted to represent a broader set of desired behaviors. This made it more flexible, but also more difficult to optimize. Because multiple objectives were combined, the agent may have received

competing learning signals during training. For example, a behavior that helped the agent stay close to the ball might not always be the same behavior that created the best offensive opportunity. Similarly, rewarding several intermediate behaviors can make it harder for the model to determine which actions are most important for long-term success.

A key limitation of the balanced reward function is that it also did not include an explicit reward for scoring goals. Although the reward structure encouraged behaviors that could contribute to successful gameplay, it did not directly optimize the agent for the main objective of Rocket League. As a result, the agent could increase its training reward by performing generally useful behaviors without necessarily learning how to consistently score, defend, or win games. This is especially important for the balanced model because combining many proxy rewards may create the appearance of well-rounded gameplay while still leaving the true game objective underrepresented.

The more aggressive balanced variant used the same general structure as the baseline balanced reward function but increased the reward weights associated with the desired behaviors. This was intended to test whether stronger reinforcement would improve learning across multiple gameplay objectives. The balanced reward function therefore served as a test of whether combining several proxy objectives could produce more general gameplay behavior, while also demonstrating the difficulty of aligning multi-objective reward shaping with the true goal of winning matches.

4 Results

The baseline and aggressive reward-function variants were trained for approximately 50 million timesteps and compared across the three reward approaches: offensive, ball-chasing, and balanced. Because the aggressive variants used larger reward weights, the raw reward values should be interpreted as shaped training performance rather than direct gameplay skill.

The resulting training logs contained approximately 7,000 recorded iterations, which were summarized to compare overall reward trends, late-stage performance, and training stability.

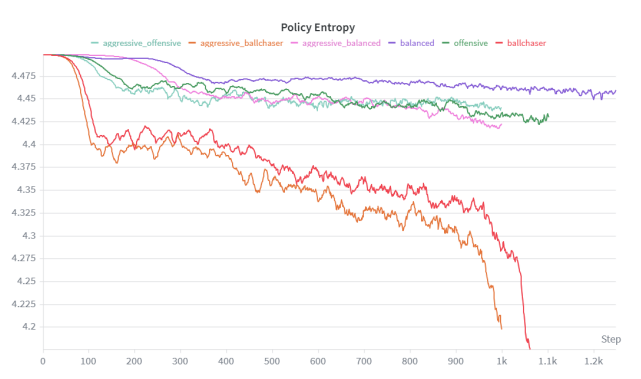


Figure 2: Policy entropy across baseline and aggressive reward-function variants.

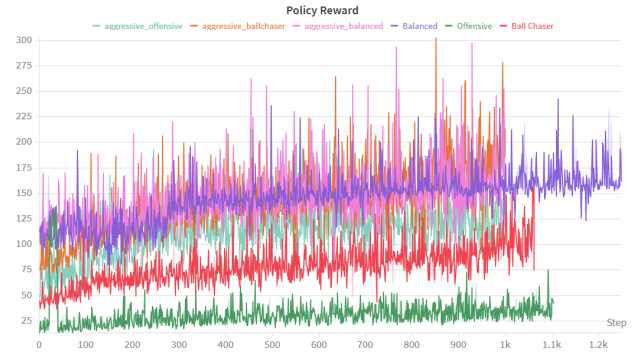


Figure 3: Policy reward across baseline and aggressive reward-function variants

4.1 Overall Training Performance

Across the full training run, the aggressive offensive and aggressive ball-chasing agents achieved substantially higher average policy rewards than their baseline counterparts. The baseline ball-chasing agent produced an average reward of 79.28, while the aggressive ball-chasing agent produced an average reward of 142.55. This was an increase of 63.27 reward points. The offensive agent showed an even larger difference, with the baseline offensive agent averaging 30.76 and the aggressive offensive agent averaging 111.67, an increase of 80.92 reward points.

The balanced reward function showed a different pattern. The baseline balanced agent achieved an average reward of 146.40, while the aggressive balanced agent averaged 145.30. This small decrease of 1.10 reward points suggests that increasing the reward weights did not meaningfully improve the balanced agent over the full training run.

Table 1: Overall Training performance

Reward	Base Mean Reward	Aggr Mean Reward	Difference
Ballchaser	79.28	142.55	+63.27
Offensive	30.76	111.67	+80.92
Balanced	146.40	145.30	-1.10

These results suggest that aggressive reward shaping was most effective for specialized behavior patterns. The offensive and ball-chasing agents each had a clearly defined behavioral goal, which may have made it easier for the agent to respond to stronger reward signals. In contrast, the balanced reward function combined several objectives, which may have reduced the benefit of simply increasing reward weights.

4.2 Late-Stage Training Performance

To evaluate end-of-training behavior, the average reward over the final 50 iterations was also compared. This metric is useful because a single final reward value can be noisy, while the final 50-iteration average gives a better sense of late-stage stability.

The aggressive ball-chasing agent had a final-50 average reward of 184.09, compared with 107.72 for the baseline ball-chasing agent. The aggressive offensive agent also showed a large improvement, averaging 123.56 over the final 50 iterations compared with 38.46 for the baseline offensive agent. These results show that the aggressive offensive and ball-chasing variants did not only improve early in training; they remained stronger near the end of the training run.

Table 2: Late Stage Performance

Reward	Base avg.	Aggressive avg.	Difference
Offensive	38.46	123.56	+85.10
Ballchaser	107.72	184.09	+76.37
Balanced	168.07	164.78	-3.29

The balanced agent again showed little benefit from the aggressive reward variant. The baseline balanced agent averaged 168.07 over the final 50 iterations, while the aggressive balanced agent averaged 164.78. This suggests that the aggressive balanced reward function did not produce a stronger late-stage policy than the baseline balanced reward function.

4.3 Peak Reward and Training Instability

All three aggressive variants achieved higher peak rewards than their baseline versions. The aggressive ball-chasing agent reached a peak reward of 302.73, compared with 156.45 for the baseline ball-chasing agent. The aggressive offensive agent reached 216.53, compared with 75.29 for the baseline offensive agent. The aggressive balanced agent also reached a high peak reward of 297.63, compared with 242.79 for the baseline balanced agent.

Table 3: Peak Reward by Agent Type

Agent Type	Baseline Peak Reward	Aggressive Peak Reward
Ball-chasing	156.45	302.73
Offensive	75.29	216.53
Balanced	242.79	297.63

However, the aggressive variants also showed larger drops from peak reward to final reward. This was especially noticeable for the aggressive balanced agent, which reached a peak reward of 297.63 but ended at 148.10, a drop of 149.54 reward points. The aggressive ball-chasing and offensive agents also declined from their peaks, but their final and final 50 average rewards still remained well above their baseline versions.

Table 4: Aggressive Agent Performance Metrics

Agent Type	Peak Reward	Final Reward	Drop from Peak
Ball-chasing	302.73	213.61	-89.11
Offensive	216.53	128.96	-87.56
Balanced	297.63	148.10	-149.54

This suggests that aggressive reward shaping can produce higher maximum reward values, but may also introduce more volatility. The balanced variant appears to be the clearest example of this tradeoff. Although it reached a high peak reward, it did not maintain that performance through the end of training.

4.4 Interpretation

The results indicate that aggressive reward shaping improved the specialized reward functions more than the general balanced reward function. The ball-chasing and offensive reward functions each focused on a narrower behavior pattern, so increasing the reward weights may have strengthened the learning signal in a useful way. The balanced reward function combined multiple objectives, so increasing the reward weights may have amplified conflicting incentives rather than improving overall gameplay behavior.

Another important consideration is that these results are based on shaped policy reward rather than direct match performance. Since the aggressive variants used higher reward weights, higher reward values do not automatically prove that the agents became better Rocket League players. Instead, the results show that the agents became better at optimizing their assigned reward functions. A stronger evaluation would require shared gameplay metrics such as goals scored, goals conceded, shots, saves, touches, win rate, or performance against fixed opponents.

5 Limitations and Future Work

This project has several limitations that should be considered when interpreting the results. First, the comparison relied primarily on shaped policy reward from the training logs. Because each reward function defines success differently, higher reward does not necessarily mean that an agent became better at Rocket League overall. This is especially important when comparing the baseline and aggressive reward variants because the aggressive versions used larger reward weights. As a result, some of the increase in policy reward may reflect reward scaling rather than true gameplay improvement.

Second, the reward functions used proxy behaviors rather than directly optimizing for match outcomes. The offensive reward function encouraged behaviors associated with attacking play, the ball-chasing reward function encouraged frequent interaction with the ball, and the balanced reward function combined several intermediate objectives. However, none of the reward functions directly rewarded scoring goals. This means the agents may have learned behaviors that appeared useful, such as chasing the ball or moving aggressively forward, without necessarily learning how to win games consistently.

Another limitation is that each reward function was tested under a single training setup. The results may depend on the specific PPO configuration, environment settings, random initialization, and training duration used in this project. Reinforcement learning is sensitive to randomness and hyperparameters, so the results should not be interpreted as universal proof that aggressive reward shaping is always better for specialized agents. Repeating the experiments with multiple random seeds would provide a more reliable estimate of each reward function’s performance.

Future work should begin by adding an explicit reward for scoring goals and a penalty for conceding goals. This would better align

the reward function with the true objective of Rocket League. The existing proxy rewards could still be used, but they should support the main game objective rather than replace it. For example, ball touches, forward pressure, and offensive positioning could receive smaller shaping rewards, while goals and conceded goals receive the largest rewards.

Additional future work could explore more advanced reward tuning. Instead of increasing the weights of the same reward components, future experiments could test different reward balances, curriculum learning, or adaptive reward schedules. For example, the agent could initially receive stronger rewards for basic behaviors such as touching the ball and moving toward the ball, then gradually shift toward rewards for shots, saves, goals, and match outcomes. This may reduce the risk of agents becoming stuck in simple behaviors like constant ball chasing.

Overall, this project shows that aggressive reward shaping can increase shaped reward performance, especially for specialized behaviors such as ball chasing and offensive pressure. However, future work is needed to evaluate whether these higher rewards translate into better gameplay. Adding goal-based rewards, using shared evaluation metrics, testing against fixed opponents, and repeating experiments across multiple random seeds would make the comparison more rigorous and better aligned with the true objective of Rocket League.

6 Conclusion

This project explored how different reward-function designs influence reinforcement learning behavior in an RLGym Rocket League environment. Three reward approaches were tested: offensive, ball-chasing, and balanced. Each reward function was designed to encourage a different style of play, and each was also tested using a more aggressive variant with increased reward weights. By comparing the baseline and aggressive versions over approximately 50 million timesteps, the project examined whether stronger reward signals improved learning performance or introduced additional instability.

The results showed that aggressive reward shaping was most effective for the specialized reward functions. The aggressive ball-chasing and aggressive offensive agents achieved substantially higher shaped rewards than their baseline counterparts, both across the full training run and during the final stages of training. This suggests that increasing reward weights can strengthen learning when the desired behavior is narrow and clearly defined. In these cases, the agents produced higher shaped reward values, suggesting that the stronger reward signals were more effective for narrow objectives such as ball chasing and offensive pressure. However, because the agents were not visually evaluated or tested with direct gameplay metrics, these results should be interpreted as training-log performance rather than confirmed gameplay improvement.

However, the balanced reward function produced a more complicated result. The aggressive balanced agent reached a high peak reward, but it did not clearly outperform the baseline balanced agent over the full training run or in late-stage performance. This suggests that increasing reward intensity does not automatically improve learning when the reward function contains multiple objectives. Instead, stronger rewards may amplify competing incentives, making

the agent more unstable or less consistent. This was one of the key findings of the project: aggressive reward shaping appears more useful for specialized behaviors than for broader, multi-objective reward functions.

The project also highlighted an important limitation in reward design. None of the tested reward functions directly rewarded scoring goals, which is the central objective of Rocket League. As a result, the agents were trained to optimize proxy behaviors such as ball movement, pressure, touches, and offensive positioning rather than direct match success. This means that higher shaped reward should not be interpreted as proof that an agent is better at winning games. Instead, the results show how different reward structures affected the agent’s measured training performance.

Overall, this project demonstrates both the usefulness and the risk of reward shaping in reinforcement learning. Well-designed rewards can guide an agent toward desired behaviors, but poorly aligned or overly aggressive rewards can also lead to instability or reward-seeking behavior that does not fully match the true objective of the game. Future work should add explicit goal-scoring rewards, penalties for conceding goals, shared gameplay evaluation metrics, and tournament-style testing against fixed opponents. These additions would make it possible to determine whether higher shaped training rewards translate into stronger real gameplay performance.

In conclusion, aggressive reward shaping improved the offensive and ball-chasing agents, but it was less effective for the balanced agent. The strongest takeaway is that reward intensity matters, but reward alignment matters more. For reinforcement learning agents, especially in complex environments like Rocket League, the reward function does not just measure learning, it defines what the agent believes success means.

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